SAP x Temporal workshop

Durable AI Agents for SAP

Date: 2026-02-11

INTERNAL – SAP and External Parties under NDA Only

Table of contents

Objectives....3
Hour 1: Why Durability Matters....3
Temporal as Durable Foundation....3
Hour 2: Hands-On – Local (or Server, or Cloud) Setup....5
Hour 3: Hands-On - Dispute Resolution w/ durability....5
Hour 4: Vision and Discussion....6

© 2026 SAP SE or an SAP affiliate company. All rights reserved. See Legal Notice on www.sap.com/legal-notice for use terms, disclaimers, disclosures, or restrictions related to this material.

INTERNAL – SAP and External Parties under NDA Only

This workshop introduces SAP architects to Temporal, a durable execution platform for SAP agents. We'll explore how to make AI agents resilient to crashes, timeouts, and failures, through the lens of an existing SAP use case.

Objectives

1. Build awareness of Temporal and how to use it in SAP's Office of the CTO teams.
2. Agents and applications get durability and cross-cutting services automatically, regardless of agent framework used.
3. Make an existing SAP use case durable. Test it, break it, see it recover.

Hour 1: Why Durability Matters

Introduce Temporal and share some example scenarios where lack of durability is a problem.

Problem Space

- A deep research agent crashes, causing hours of work to be lost and requiring the process be started from scratch.
- A failure in multi-agent orchestration fails, resulting in inconsistent state across agents and manual cleanup.
- A long-running task times out, no progress is preserved, resulting in wasted compute.
- Downstream A2A call fails, context is lost, and process must restart.

There is some code in our Dispute Resolution agent that can be used to demonstrate: https://github.com/SAP-samples/btp-a2a-dispute-resolution/blob/main/agents/sap-agent-builder-a2a/agent-builder-a2a-agent-connector/srv/BafAgentClient.ts#L55

Specifically, a few places in the while (true) block where a crash would potentially cause loss of context because the polling loop holds all state in memory. In this business scenario, that means a user having to explain all or part of the problem again from the start. We can demo with a local express app or the deployed agent to illustrate.

© 2026 SAP SE or an SAP affiliate company. All rights reserved. See Legal Notice on www.sap.com/legal-notice for use terms, disclaimers, disclosures, or restrictions related to this material.

INTERNAL – SAP and External Parties under NDA Only

Temporal as Durable Foundation

Core concepts about Temporal and durability

What you write	What Temporal does
Normal async function	Record of every step
await a function call	Resumable checkpoint
Process crashes	Automatically restarts from last checkpoint
External call fails	Retries with exponential backoff

A vision for cross-cutting durable services

Service	Goal
SAP Generative AI Hub	• LLM calls are automatically durable • Token counts are tracked across retries
Observability	• OTEL traces for every workflow step • Failures surfaced in central trace (e.g., Dynatrace, Jaeger, SAP Cloud Logging)
SAP HANA Cloud	Workflow state persistence

Key Insights

1. Temporal replays deterministic steps from history
2. It skips already-completed activities
3. It resumes from the exact crash point

Outcomes

1. Architects understand Temporal positioning and scenarios.
2. Background on topic of durability and why it is important for agents.
3. Set the stage with a real use case and SAP context.

Discussion: What does a durability-aware Gen AI Hub look like?

1. Retry on 429/5xx with exponential backoff

© 2026 SAP SE or an SAP affiliate company. All rights reserved. See Legal Notice on www.sap.com/legal-notice for use terms, disclaimers, disclosures, or restrictions related to this material.

INTERNAL – SAP and External Parties under NDA Only

2. Don’t double-charge for retried requests
3. Token usage aggregated across retries
4. Model fallback (e.g., Claude to GPT if one fails)

Hour 2: Hands-On – Local (or Server, or Cloud) Setup

1. Hello World exercise with Temporal (TS equivalent)
2. Temporal plugins – SAP Gen AI Hub Vercel AI SDK Adapter + LiteLLM

   Key Code Pattern (TypeScript)

```
import { createSAPAI } from '@sap/ai-sdk-vercel-adapter';
import { generateText } from 'ai';

const sapai = createSAPAI();

const result = await generateText({
  model: sapai.languageModel('anthropic--claude-4.5-sonnet'),
  prompt: 'Analyze this invoice dispute...',
});

console.log(result.text);
```

3. Durable Agent with Tools sample (TS equivalent)

Hour 3: Hands-On - Dispute Resolution w/ durability

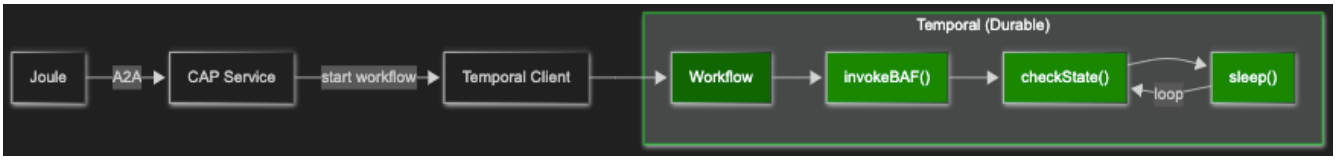
Current architecture of the Dispute Resolution agent includes a polling mechanism to SAP Business Agent Foundation (BAF). It has the following challenge:

© 2026 SAP SE or an SAP affiliate company. All rights reserved. See Legal Notice on www.sap.com/legal-notice for use terms, disclaimers, disclosures, or restrictions related to this material.

INTERNAL – SAP and External Parties under NDA Only

Problem: The polling loop (in red) holds all state in memory. A crash loses everything.

Solution: Create the functions as workflow and activity, introducing durability with Temporal. This ensures each step is recorded, enabling resume from last checkpoint.



Activities

1. Create a TS project and add temporal and SAP AI SDK Vercel Adapter
2. Define Activities
3. Define Workflows
4. Create Worker
5. Create Client
6. Test Failure Recovery
7. Watch the Magic

Hour 4: Vision and Discussion

Possible topics:

1. Service integration → How can we provide more durability via backing services out of the box?
2. Toolkit for agent development → Temporal baked in and more or less transparent to developer
3. Durability in Joule core?

www.sap.com.

© 2026 SAP SE or an SAP affiliate company. All rights reserved. See Legal Notice on www.sap.com/legal-notice for use terms, disclaimers, disclosures, or restrictions related to this material.

INTERNAL – SAP and External Parties under NDA Only